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Vertebral endplate shaver with height adjustable blades

Abstract

A vertebral endplate preparation (decorticating) tool has height adjustable blades at a distal end of a shaft assembly for decorticating/shaving vertebral endplate material from spinal vertebrae, translates axial movement of a control rod o the shaft assembly of the vertebral endplate preparation tool with height adjustable blades into radial blade height change (extension and retraction) through a controller of a handle assembly. The blades may also be configured for collecting and removing shaved vertebral endplate material. The blades controllably move outward radially from the distal end of the shaft assembly and controllably move inward radially into the distal end of the shaft assembly upon manipulation of the controller. Radial blade extension and retraction from the distal end may be accomplished via mutual angled dovetail features of the blades and the distal end, or via pins of the distal end of the shaft assembly situated in angled slots of the blades.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This U.S. non-provisional patent application claims the benefit of and/or priority under 35 U.S.C. § 119(e) to U.S. provisional patent application Ser. No. 63/281,915 filed November 22, 2021 titled “Expandable Vertebral Disc Micro Shaver” and U.S. provisional patent application Ser. No. 63/315,638 filed Mar. 2, 2022 titled “MIS Vertebral Endplate Shaver With Height Adjustable Blades,” the entire contents of each of which are specifically incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates to instruments for spine procedures and, more particularly, to medical instruments for removing vertebral endplate material from a vertebra in a vertebral disc

space of a spine.

BACKGROUND OF THE INVENTION

(2) Many people contend with spine issues due to age, disease, trauma, congenital, and acquired complications and conditions. While some spine issues can be alleviated without surgery, other spine issues necessitate surgery. Spine surgery may entail removing vertebral disc material from between adjacent vertebrae. This can be accomplished using minimally invasive surgery, micro surgery, or the like. Such surgery reduces trauma by using surgical instruments that are introduced into the body via one or more small incisions. Certain of these medical instruments are introduced into the body via an access tube, endoscope, cannula or the like (collectively, cannula) that has been inserted into the body through a small incision and positioned accordingly. Other medical instruments can then be inserted into the cannula and controlled accordingly. Some medical instruments include a “built-in” cannula wherein the medical instrument with the cannula is inserted into an incision, positioned accordingly, and manipulated.

(3) Various surgical spine procedures may be performed via a cannula (separate from or as part of the medical instrument). When an intervertebral spine implant is to be installed between adjacent vertebrae (i.e. in the vertebral disc space) it is necessary to properly prepare endplate surfaces of the adjacent vertebrae. Proper vertebral endplate surface preparation typically includes using a medical instrument with a cannula such as a vertebral endplate preparation instrument/tool (vertebral endplate shaver) to decorticate or shave the vertebral endplate surfaces to ensure adequate seating of the intervertebral spine implant. Typical vertebral endplate preparation tools have static blades (i.e. their blade(s) are fixed in height) that are inserted into a vertebral disc space and rotated to decorticate vertebral endplate surfaces of adjacent vertebrae. Because the blades are fixed in height a series of static vertebral endplate preparation tools each one having blades of a different radial height is therefore used to decorticate or shave the vertebral endplate surfaces. Such a series may be deemed a set of static vertebral endplate preparation tools.

(4) Using static vertebral endplate preparation tools to properly decorticate vertebral endplate surfaces can increase the likelihood of problems since multiple static vertebral endplate preparation tools must be inserted into and removed from the cannula during the spine procedure. However, with a vertebral endplate preparation tool having height adjustable (dynamic) blades, a single medical instrument can be used to decorticate vertebral endplate surfaces. It is therefore preferable to use a vertebral endplate preparation tool having blades that can be easily varied in height.

(5) Various vertebral endplate preparation tools with height adjustable blades have heretofore been devised. However, prior art vertebral endplate preparation tools with height adjustable blades suffer from several shortcomings such as, but not limited to, overall ease of use, ease of varying blade height, and not being conducive to use with a cannula.

(6) It would therefore be desirable to have a vertebral endplate preparation tool with height adjustable (dynamic) blades that overcomes the deficiencies of prior art vertebral endplate preparation tools with height adjustable blades. The present vertebral endplate preparation tool with height adjustable (dynamic) blades addresses the above and more.

SUMMARY OF THE INVENTION

(7) A medical instrument in the form of a vertebral endplate preparation tool with height adjustable (dynamic) blades for decorticating/shaving vertebral endplates during a spine procedure particularly, but not necessarily, using a cannula, has a head with at least one height adjustable blade for decorticating/shaving vertebral endplate material from a vertebra of a spine. The height adjustable blade(s) is/are configured to controllably move outward radially from the head and controllably move inward radially into the head upon manipulation of a controller of the vertebral endplate preparation tool with height adjustable instrument that is coupled to the head. The amount of outward radial movement of the blade(s) is controllable to set a height of the blade(s). With the blade(s) radially moved outward, they extend radially beyond a diameter of the head and an associated cannula. Rotation of the head via a handle of the vertebral endplate preparation tool with

height adjustable rotates the blade(s) to allow decorticating of the vertebral endplates.

(8) In one form, the present vertebral endplate preparation tool with height adjustable blades has a handle, a controller operably connected to the handle, a control shaft defining a distal control shaft end and a proximal control shaft end that is operably connected to the controller for controlled manipulation of the control shaft, and a head on the distal control shaft end having first and second blades for decorticating or shaving vertebral endplates of vertebrae of a spine. The first and second blades controllably move outward radially (expand) from the head, and controllably move inward radially (collapse) into the head upon manipulation of the controller. The vertebral endplate preparation tool with height adjustable blades may include a cannula with the control shaft disposed within. Movement of the handle operates the controller to manipulate the control shaft to axially extend the head from a cannula, controllably expand the first and second blades radially outward to a desired height from the head beyond a diameter of the cannula after the head has been extended from the cannula, and contract the first and second blades into the head when vertebral endplate decorticating/shaving is finished.

(9) In one form, each blade has a concave portion for allowing shaved vertebral endplate material to collect therein. In one form, cutouts or windows in a lower portion of the blade(s) aid in allowing vertebral endplate material to be collected, then removed. When rotated, the blades(s) provide an almost 180° collection sweep outside of a longitudinal plane (axis) of the associated cannula.

(10) The head has a nose and a rear or base that each slidingly hold an end of each blade for radial expansion/contraction of each blade from the head. The moving connection between the nose and an end of the blades is preferably, but not necessarily, via angled dovetail configurations. The moving connection between the base and another end of the blades is also preferably, but not necessarily, via angled dovetail configurations.

(11) Once the head is external of the cannula, axial movement in one direction of a component of the control shaft that is connected to a nose of the head, pulls the nose axially towards the base of the head to provide axial compression. Axial compression of the nose towards the base axially compresses against the movable blades to slide them radially outward from the nose and base to expand the blades from the head. Axial movement of the control shaft in an axially opposite direction allows retraction of the blades into the head, through release of axial pressure.

(12) In another form, the dynamic vertebral endplate preparation tool has height adjustable blades at a distal end of a shaft assembly for decorticating/shaving vertebral endplate material from vertebrae of a spine, translates axial movement of a control rod of a shaft assembly into radial blade height change (radial blade height extension or retraction) through a controller of a handle assembly of the dynamic vertebral endplate preparation tool. The blades controllably move outward radially (extend) from the distal end of the shaft assembly and controllably move inward radially (retract) into the distal end of the shaft assembly upon manipulation of the controller in a head of the dynamic vertebral endplate preparation tool that is connected to the shaft assembly.

(13) Once the blades are expanded, they extend radially beyond a diameter of an associated cannula. Rotation of the blades turn the blades to provide decorticating/shaving of vertebral endplate material. The blades may be configured to also collect and removal of shaven vertebral endplate material.

(14) Radial extension and retraction of the blades from the distal end of the shaft assembly is accomplished in one form via mutual angled dovetail features of the blades and the distal end whereby axial force against the blades (compression) via the control rod effects radially outward movement (extension) of the blades via the mutual angled dovetail features between the blades and the distal end (a first axial direction). Reverse axial movement of the control rod releases axial pressure (force) against the blades to effect radially inward movement (retraction) of the blades. In one form, the distal end has angled dovetail slots or channels that receive angled dovetail ends or flanges of the blades (a second axial direction opposite the first axial direction). In a particular

form, the distal end has front angled dovetail slots/channels and rear angled dovetail slots/channels that receive respectfully a front dovetail end or flange and a rear dovetail end or flange of each blade.
(15) Radial extension and retraction of the blades from the distal end is accomplished in one form via pins of the distal end situated in angled slots of the blades whereby axial translation of the pins against the angled slots of the blades in one direction effects radially outward movement (extension) of the blades, while axial translation of the pins against the angled slots of the blades in the reverse direction effects radially inward movement (retraction) of the blades. In a particular form, each blade has two angled slots while the control rod has two pins, one pin for each of the two angled slots of the blades. The two angled slots of each blade are situated skewed relative to each other.
(16) In use, manipulation of the controller of the handle provides extension of the blades radially outward from the distal end beyond a diameter of the cannula after the distal end is clear of the cannula, and retract the blades into the distal end when use of the dynamic vertebral endplate preparation tool is finished.
(17) Each blade may include a trough for collecting, and then removing, the shaven vertebral endplate material. In a particular form, cutouts or windows in a lower portion of the blade(s) aid in collection of shaved vertebral endplate material during use. When rotated, the blades(s) provide an almost 180° collection sweep outside of a longitudinal plane (axis) of the associated cannula.
(18) Further aspects of the present invention will become apparent from consideration of the drawings and the following description of forms of the invention. A person skilled in the art will realize that other forms of the invention are possible and that the details of the invention can be modified in a number of respects without departing from the inventive concept. The following drawings and description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention and its features will be better understood by reference to the accompanying drawings, wherein:
(2) FIG. **1** is a view of an exemplary vertebral endplate preparation tool with height adjustable blades for decorticating/shaving vertebra endplate material from vertebrae of a spine fashioned in accordance with the present principles;
(3) FIG. **2** is an enlarged view of a middle to distal portion of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1**, depicting a bladed head thereof ready for extension and deployment from a cannula;
(4) FIG. **3** is an enlarged view of a distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head shown in an extended but un-deployed position;
(5) FIG. **4** is an enlarged view the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head shown in an extended and deployed position;
(6) FIG. **5** is an enlarged view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head shown in an extended, deployed and rotated position;
(7) FIG. **6** is an enlarged side view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head shown in an extended but un-deployed position;
(8) FIG. **7** is an enlarged side view of the distal end of the exemplary vertebral endplate preparation

tool with height adjustable blades of FIG. **1** with the bladed head in an extended and deployed position;
(9) FIG. **8** is an enlarged view of a controller of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1**;
(10) FIG. **9** is another enlarged view of the controller of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1**;
(11) FIG. **10** is an enlarged side view of the bladed head and distal end portion of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head in an un-deployed position;
(12) FIG. **11** is an enlarged view of the bladed head and distal end portion of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head and distal end portion in an un-deployed position;
(13) FIG. **12** is an enlarged side view of the bladed head and distal end portion of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head and distal end portion in a deployed position;
(14) FIG. **13** is an enlarged side view of the bladed head and distal end portion of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head and distal end portion in an un-deployed position;
(15) FIG. **14** is another enlarged side view of the bladed head and distal end portion of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **1** with the bladed head and distal end portion in an un-deployed position;
(16) FIG. **15** is a view of another exemplary vertebral endplate preparation tool with height adjustable blades for decorticating/shaving vertebra endplate material from vertebrae of a spine fashioned in accordance with the present principles;
(17) FIG. **16** is an enlarged view of a proximal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **15**, particularly showing the handle assembly;
(18) FIG. **17** is an enlarged view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **15** with the blades shown retracted;
(19) FIG. **18** is another enlarged view the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **15** with the blades shown retracted;
(20) FIG. **19** is an enlarged generally side view of the distal end of the exemplary with height adjustable blades vertebral endplate preparation tool of FIG. **15** with the blades shown extended;
(21) FIG. **20** is an enlarged front side view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **15** with the blades shown extended;
(22) FIG. **21** is a top view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **15** with the blades shown retracted;
(23) FIG. **22** is an enlarged side sectional view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **15** with the blades shown retracted;
(24) FIG. **23** is another enlarged side sectional view of the distal end of the exemplary vertebral endplate preparation tool of with height adjustable blades FIG. **15** with the blades shown extended;
(25) FIG. **24** is an enlarged view of a distal end of a further exemplary form of a vertebral endplate preparation tool with height adjustable blades for decorticating/shaving vertebral endplate material from vertebrae of a spine fashioned in accordance with the present principles with the blades thereof shown extended;
(26) FIG. **25** is an enlarged view of a distal end of a still further exemplary form of a vertebral endplate preparation tool with height adjustable blades for decorticating/shaving vertebra endplate material from vertebrae of a spine fashioned in accordance with the present principles with blades of the distal end shown retracted;
(27) FIG. **26** is an enlarged side view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **25** with the blades shown extended; and

(28) FIG. **27** is an enlarged side sectional view of the distal end of the exemplary vertebral endplate preparation tool with height adjustable blades of FIG. **25** with the blades shown extended.
(29) For the purposes of promoting an understanding of the principles of the invention reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Any alterations and further modifications in the described embodiments, and any further applications of the principles of the invention as described herein are contemplated as would normally occur to one skilled in the art to which the invention relates.
DETAILED DESCRIPTION OF THE INVENTION
(30) Referring to FIGS. **1-14**, there is shown an exemplary medical instrument, generally designated **10** for decorticating or shaving vertebral endplate material from vertebrae of a spine, and optionally collecting and removing vertebral endplate material (not shown) from a vertebra (not shown) of a spine (not shown) during a surgical spine procedure, particularly, but not necessarily, using a cannula. The medical instrument **10** may be considered a vertebral endplate preparation tool with height adjustable (dynamic) blades and is made from one or more surgical grade materials. The vertebral endplate preparation tool with height adjustable blades **10** has a head **15** with blades **22**, **24** designed to shave, collect, and remove vertebral disc material from a vertebral disc. The blades or shavers **22**, **24** are movably retained in the head **15** such that each blade **22**, **24** is able to extend or spread outward radially from the head, and retract or collapse inward radially from the head when actuated by the vertebral endplate preparation tool with height adjustable blades **10**. The radial spread of the blades **22**, **24** is beyond the diameter of an associated cannula **80**, such as, but not necessarily, to the height of a vertebral disc space. Axial compression of the nose **20** to the base **46** of the head **15** that slidingly retain curved ends **50**, **51** of the blade **22** and curved ends **48**, **49** of the blade **24** provides radial expansion of the blades **22**, **24**, while relieving axial compression provides radial contraction of the blades **22**, **24**.
(31) FIG. **1** depicts an overall view of the vertebral endplate preparation tool with height adjustable blades **10**. The vertebral endplate preparation tool with height adjustable blades **10** has a handle **12**, a controller **13** operably coupled to the handle **12**, a control shaft **14** operably coupled to the controller **14**, and a head **15** operably coupled to a distal end **18** of the control shaft **14**. The handle **12** is characterized by a gently curved body **16** sized to accommodate a hand (not shown) and includes several indentions **17***a*, **17***b*, **17***c*, **17***d* each configured to accommodate a finger. Other configurations may be used.
(32) FIG. **2** depicts a cannula construction **78** defined by a hollow longitudinal cannula **80** having a distal angled end **81**, an inner longitudinal bore **82**, and a proximal end **83** that is shown receiving a secondary cannula **85** having a receiving end **84**. The control shaft **14** is shown extended into the cannula construction **78** with the head **15** of the medical instrument **10** in a pre or post-deployment/expansion position at the distal angled end **81** of the cannula **80**. Through the use of telescoping cannula construction **78**, overall length of the cannula may be adjusted as desired. Other configurations may be used.
(33) FIGS. **3-7** depict various views of the head **15** of the present vertebral endplate preparation tool with height adjustable blades **10** as it emerges from the end **81** of bore **82** of the cannula **80**. The overall shape of the head **15** is cylindrical which therefore defines arcs or arcuate portions about the circumference and/or the axial length of the overall cylindrical shape of the head. The head **15** has a nose **20** at a distal end of the head **15**, and a base **46** at a proximal end of the head **15**. The nose **20** is shaped generally as a flattened arch, but other shapes may be used. The base **46** is fixed to the control shaft **14**, which itself axially moves within the cannula **80**, as regulated by the controller **13** via the handle **12**. As seen in FIGS. **3**, **4**, and **6**, for example, the base **46** is situated at and fixed to the distal end of the control shaft **14**. A first blade or shaver **22** is situated axially between the nose **20** and the base **46** along an axial length of the outer diameter of the cylindrical shape of the head **15**, while a second blade or shaver **24** is situated axially between the nose **20** and

the base **46** along another axial length of the outer diameter of the cylindrical shape of the head **15** and generally diametrically opposite the first blade **22**, the nomenclature first and second being arbitrary. The first blade **22** is held between the nose **20** and the base **46** such that the first blade **22** can move radially outward (expand) and radially inward (collapse) with respect to the head **15**. The first blade **22** further includes a concavity, groove, cutout, depression, scoop, or the like **26** that extends along the curved longitudinal edge of the first blade **22** which is configured to receive and/or collect shaved vertebral disc material. A window **23** is provided in the first blade **22** that aids in collection of shaved vertebral disc material. The second blade **24** is also held between the nose **20** and the base **46** such that the second blade **24** can move radially outward (expand) and radially inward (collapse) with respect to the head **15**. The second blade **24** further includes a concavity, groove, cutout, depression, scoop, or the like (not seen) that extends along the curved longitudinal edge of the second blade **24**, which is configured to receive and/or collect shaved vertebral disc material in like manner and form as scoop **26**. A window **25** is provided in the second blade **24** that aids in collection of shaved vertebral disc material. Axial motion is controlled through the control shaft **14** from the controller **13** through the handle **14**.

(34) The nose **20** has a collar or skirt **35** (collectively, collar) that extends axially from the proximal end of the nose **20** a distance towards the base **46**. One side of the collar **35** has a first, preferably, but not necessarily, angled slot **40** that is configured to movably or slidingly receive a first angled side **50** of the blade **22**. The first slot **40** of the nose **20** and the first angled side **50** of the blade **22** are preferably, but not necessarily, formed between them as a dovetail joint. Another side of the collar **35** has a second, preferably, but not necessarily, angled slot **41** that is configured to movably or slidingly receive a first angled side **48** of the second blade **24**. The second slot **41** of the nose **20** and the first angled side **48** of the second blade **24** are preferably, but not necessarily, formed between them as a dovetail joint.

(35) The base **46** has a first, preferably, but not necessarily, angled slot **44** that is configured to movably or slidingly receive a second angled side **51** of the blade **22**. The first slot **44** of the base **46** and the second angled side **51** of the blade **22** are preferably, but not necessarily, formed between them as a dual angled dovetail joint/channels. Another side of the base **46** has a second, preferably, but not necessarily, angled slot **45** that is configured to movably or slidingly receive a second angled side **49** of the second blade **24**. The second slot **44** of the base **46** and the second angled side **49** of the second blade **24** are preferably, but not necessarily, formed between them as a dual angled dovetail joint/channels. Through axial compression and relief of axial pressure of the nose **20** axially towards the base **46** and axially away from the base **46**, the first and second blades **22**, **24** are caused to move relative to the nose **20** and the base **46** because of the slanted ends of the blades and the slots of the nose and the base. Axial pressure of the nose **20** towards the base **46** causes the first and second blades **22**, **24** to slide up, out and along (spread out or expand from the head **15**) the dovetail configurations between the ends of the blades and the slots of the nose and base, while relief of axial pressure of the nose **20** away from the base **46** causes the first and second blades **22**, **24** to slide down, in and along (collapse or compress into the head **15**) the dovetail configurations between the ends of the blade and the slots of the nose and the base.

(36) The head **15** also has an actuator **30** that extends from the collar **35** of the nose **20** towards the base **46**. The actuator **30** provides a connection for controlling axial movement of the nose **20** relative to the base **46** for controlling blade expansion/contraction. While not seen in the figures, the head **15** may include a second actuator situated opposite to the actuator **30** that would mimic the actuator **30**. The actuator **30** includes a first arm **31** that extends axially from the end of the collar **35** and is connected to one side of a first plate **32**. A second arm **33** extends axially from a second side of the first plate **32** towards and through a channel **47** in the side of the base **46**. The second arm **33** further extends axially through and is retained in a groove **34** formed in an outside surface of the control shaft **14**. As signified by the double-headed arrow in FIGS. **3**, **4** and **6**, the actuator **30** is able to axially translate relative to the base **46** and control shaft **14**. This is

accomplished by a translational shaft **64** of the control shaft **14** that is connected to the actuator **30** and which can axially move relative to the body of the control shaft **14** through the controller **13**. When the translational shaft **64** is axially translated in one direction, the actuator **30** axially pulls against the nose **20** to squeeze (axially translate) it towards the base **46** thereby pushing out (expanding) the first and second blades **22**, **24**. Axial translation of the translational shaft **64** in the opposite axial direction pushes the nose **20** axially away from the base **64** to collapse the first and second blades **22**, **24** into the head **15**.

(37) FIGS. **3-5** depict expansion of the blades **22**, **24** from the head **15** as the head **15** emerges (is axially translated from) the cannula **80**. FIG. **6** depicts a side view of the head **15** with the blades **22**, **24** un-expanded, while FIG. **7** depicts a side view of the head **15** with the blades **22**, **24** expanded. FIGS. **10-14** depict various views of the head **15** with the blade **22**, **24** expanded and un-expanded showing the control shaft **14** and the translational shaft **64** relative to the actuator **30** and blades **22**, **24**.

(38) FIGS. **8-9** depict two views of the controller **13**. The controller **13** has a body **56** connected to the handle **12** via a neck **58**. The handle **12** is operably coupled to a control knob **60** for axially moving the control shaft **14** within the cannula **80**. A toothed tab **59** is slidingly retained by the body **13** and when slid into a locked position (FIG. **9**) where the teeth thereof meet end **61** of the control knob **60**. Thus, control knob **60** is locked in orientation via sliding the lock tab **59** down from the open position (shown in FIG. **8**) to a locked position (shown in FIG. **9**) which will prevent (e.g., lock) the blades **22**, **24** from collapsing into the head **15** when in use.

(39) The cannula **80** extends from the neck **57** of the body **56** of the controller **13** and may include a head position indicator **52** for indicating status of the head **15**. The indicator **52** includes an opening or window **53** formed in the cannula **80** with demarcations **54** about the opening **53** to indicate axial position of the head **15** relative to the opening **81** of the cannula **80**. The control shaft **14** has a mark **55** that will move axially within the opening **53** with axial movement of the control shaft **14**, as signified by the double-headed arrow in FIG. **8**.

(40) Referring to FIGS. **15-23**, there is shown another form of a vertebral endplate preparation tool with height adjustable blades generally designated **110** for decorticating/shaving vertebral endplate material from vertebrae (not shown) of a spine (not shown) during a surgical procedure. The medical instrument **110** may also be configured and used for collecting and removing the shaved vertebral endplate material. The medical instrument **110** is preferably, but not necessarily, used with a cannula or endoscope during a minimally invasive or micro invasive spine procedure. The vertebral endplate preparation tool with height adjustable blades **110** is made from one or more surgical grade materials. In general, the vertebral endplate preparation tool with height adjustable blades **110** has a proximal end/end assembly **111** with a handle **112** that is connected to a proximal end of a controller **113** via a neck **118**, a shaft assembly **114** connected at a proximal end thereof to a distal end of the controller **113** via a collar **115**, and a head **122** at a distal end **120** of the shaft assembly **114**, the head **122** having a nose **121** and a blade assemblage/assembly **133** having first and second blades **136**, **137** designed to decorticate or shave vertebral endplate material from a vertebra (not shown), the nomenclature first and second being arbitrary here and throughout if not specifically indicated otherwise. The vertebral endplate preparation tool with height adjustable blades **1110** may also be configured to collect and remove the shaved vertebral endplate material and is so configured as shown. The first and second blades **136**, **137** are movably retained in the head **122** such that each blade **136**, **137** is able to extend or spread outward radially from the head **122**, and retract or collapse inward radially from the head **122** when actuated by the vertebral endplate preparation tool with height adjustable blades **1110**. The radial spread of the first and second blades **136**, **137** is beyond the diameter of an associated cannula, access tube, endoscope or the like, such as, but not necessarily, to the height of a vertebral disc space (not shown), such that the medical instrument **1110** may decorticate or shave vertebral endplate material from adjacent vertebra within the disc space if desired, rather than each vertebral endplate of adjacent vertebrae

separately.

(41) Referring to FIG. **16**, the handle **112** is characterized by a gently curved body **116** sized to accommodate a hand (not shown) and includes several indentions **117***a*, **117***b*, **117***c*, **117***d* each configured to accommodate a finger. Other configurations may be used. The controller **113** has a frame defined by upper and lower frame members **128***a*, **128***b* that project from the neck **118** of the handle **112**. The upper and lower frame members **128***a*, **128***b* are connected to a front frame member **129** that connects to the collar **115**. The controller **113** has a rotatable control knob **130** that adjusts the extension and contraction of the first and second blades **136**, **137** through the handle **112** that is operably coupled to the control knob **160** for axially moving the control shaft **114** within the cannula **180**. A toothed tab **175** is slidingly retained by the controller **113**. In FIG. **16**, the toothed tab **175** is in an unlocked position axially away from the rotatable control knob **160** such that the toothed tab **175** is not engaged with the rotatable control knob **160**. When the toothed tab **175** is axially slid into a locked position (not shown) where the teeth thereof meet the toothed end **174** of the rotatable control knob **160**, rotation of the handle **112** rotates the toothed tab **175** which rotates the rotatable control knob **160** to effect blade extension and retraction through the shaft assembly **114**.

(42) The shaft assembly **114** includes a generally hollow, longitudinal tube **119** having upper and lower flats along its longitudinal length and an internal control rod (not seen) that extends at a proximal end thereof from the rotatable control knob **160** to and operably connected with the blade assemblage **133** of the head **122** at a distal. The proximal end of the internal control rod is externally threaded. The rotatable control knob **160** has an internally threaded bore (not seen) that receives the externally threaded proximal end of the internal control rod of the shaft assembly **114**. Rotation of the rotatable control knob **160** axially moves the internal control rod of the shaft assembly **114**, the direction of axial movement of the internal control rod depending on the rotational direction of the rotatable control knob **160** through threaded engagement. As explained further below, axial movement of the internal control rod effects extension and retraction of the first and second blades **136**, **137** from the head **122**. The longitudinal tube **119** includes a blade height indicator **124** that provides a visual indication of blade height from the head **122**. The height indicator **124** has an opening or window **125** formed in the longitudinal tube **119** with demarcations **126** about the opening **125** to indicate height position of the first and second blades **136**, **137** from the head **122**. The internal control rod **114** has a marker **126** that will move axially within the opening **125** with axial movement of the internal control rod.

(43) FIGS. **17-21** depict various views of the head **122** and blade assemblage **133** of the present vertebral endplate preparation tool with height adjustable blades **110**. The overall shape of the head **122** is generally cylindrical. The head **122** has a nose **121** at a distal end thereof, with a first surface **159** on a first lateral side, and a second surface **171** on a second lateral side. The first lateral surface **159** has a first arm **158** extending axially to the distal end of the longitudinal tube **119** with a first rear tail **160** received in a first axial slot **156** of the distal end of the longitudinal tube **119**. The second lateral surface **171** has a second arm **170** extending axially to the distal end of the longitudinal tube **119** with a second rear tail **172** received in a second axial slot **168** of the distal end of the longitudinal tube **119**, the first and second arms **158**, **170** and the first and second rear tails **160**, **172** being preferably, but not necessarily, opposite one another. The first and second arms **158**, **170** and thus the nose **121** of the head **122** is longitudinally axially movable relative to the longitudinal tube **119**, since the internal control rod of the shaft assembly **114** is attached thereto. Axial movement of the internal control rod towards the controller **113** pulls the nose **121** and arms **158**, **170** to compress against the first and second blades **136**, **137** thereby effecting blade extension through mutual dovetail features of the first and second blades **136**, **137** and the distal end **122** along with slanted first and second channels **143**, **152** of the first and second blades **136**, **137** respectively, wherein the amount of axial movement determines radial height (extension) of the first and second blades **136**, **137**. Axial movement of the internal control rod of the shaft assembly

114 away from the controller **113** pushes the nose **121** and arms **158**, **170** releases compression against the first and second blades **136**, **137** to effect retraction of the first and second blades **136**, **137**.

(44) The first blade **136** is situated axially between the nose **121** and the distal end of the longitudinal tube **119** of the shaft assembly **114**. The first blade **136** is generally trapezoidal in overall shape with a gently curved outer longitudinal edge. As best discerned in FIGS. **19** and **20**, the first blade **136** has a first slanted front **138** with at least a generally dovetail configuration and a first slanted rear **139** with at least a generally dovetail configuration. The gently curved outer longitudinal edge of the first blade **136** has a groove or channel **141** that extends between a first front end **140** and a first rear end **142** that is configured to receive and/or collect shaved vertebral endplate material. The groove **141** is on one side of the gently curved outer longitudinal edge of the first blade **136**, while the opposite side of the gently curved outer longitudinal edge does not have a groove. As such, the first blade **136** is configured to be rotated in one direction (here, a clockwise direction) in order to shave vertebral endplate material from a vertebra (not shown). A first front cutout **144** and a first rear cutout **145** is provided in the first blade **136** that aid in collection of shaved vertebral endplate material. Situated between the first front cutout **144** and the first rear cutout **145** is a first angled slot **143**. As best discerned in FIGS. **22** and **23**, the first angled slot **143** is slanted from a distal position to the proximal direction. The first angled slot **143** is sized to receive a pin **162** that is held between and extends from the first arm **158** to the second arm **170**. A first spacer **161** is provided between the first arm **158** and the first blade **136**.

(45) The second blade **137** is situated axially between the nose **121** and the distal end of the longitudinal tube of the shaft assembly **114**. The second blade **137** is generally trapezoidal in overall shape with a gently curved outer longitudinal edge. As best discerned in FIGS. **19** and **20**, the second blade **137** has a second slanted front **148** with at least a generally dovetail configuration and a second slanted rear **149** with at least a generally dovetail configuration. The gently curved outer longitudinal edge of the second blade **137** has a groove or channel (not seen) that extends between a second front end (not seen) similar to the first front end **140** of the first blade **136**, and a second rear end (not seen) similar to the first rear end **142** of the first blade **136**, that is configured to receive and/or collect shaved vertebral endplate material. The groove (not seen) is on one side of the gently curved outer longitudinal edge of the second blade **137**, while the opposite side of the gently curved outer longitudinal edge does not have a groove. The groove (not seen) of the second blade **137** is on the side of the second blade **137** such that rotation in the same direction as the first blade **136** (here, a clockwise direction) provides shaving of vertebral endplate material from a vertebra (not shown). A second front cutout **153** and a second rear cutout **154** is provided in the second blade **137** that aid in collection of shaved vertebral endplate material. Situated between the second front cutout **153** and the second rear cutout **154** is a second angled slot **152**. The second angled slot **152** is slanted from a distal position to the proximal direction in like manner to the first angled slot **143**. The second angled slot **152** is sized to receive the pin **162** that is held between and extends from the first arm **158** to the second arm **170**. A second spacer **173** is provided between the first arm **158** and the second blade **137**.

(46) The proximal end of the nose **121** has a first angled lower slot **166** that is configured as a dovetail slot to movably or slidingly receive the first angled front **138** of the first blade **136**, whereby the first lower angled slot **166** and the first angled front **138** together define a dovetail joint. Other manners of connection may be used and are contemplated. The distal end of the longitudinal tube **119** has a second angled lower slot **167** that is configured as dovetail slot to movably or slidingly receive the first angled rear **139** of the first blade **136**, whereby the second angled lower slot **167** and the first angled rear **139** together define a dovetail joint. Other manners of connection may be used and are contemplated. The slant of the first angled lower slot **166**/first angled front **138** of the first blade **136** pair dovetail joint, and the slant of the second angled lower slot **167**/first angled rear **139** of the first blade **36** pair dovetail joint are opposite one another,

whereby axial compression against the first angled front **138** and the first angled rear **139** of the first blade **136** causes, at least in part, the first blade **136** to extend radially outward from the distal end **122**. The pin **162** is connected to the distal end of the internal control rod. As the pin **162** is pulled axially towards the proximal end/end assembly **111**, the first angled slot **143** of the first blade **136** rides against the pin **162** during extension. Reverse axial movement of the internal control rod pushes the pin **162** against the first angled slot **143** such that the first blade **136** is retracted. The amount of axial movement in the proximal direction determines the amount of blade extension.

(47) The proximal end of the nose **121** further has a first angled upper slot **164** that is configured as a dovetail slot to movably or slidingly receive the second angled front **148** of the second blade **137**, whereby the first angled upper slot **164** and the second angled front **148** together define a dovetail joint. Other manners of connection may be used and are contemplated. The distal end of the longitudinal tube **119** has a second angled upper slot **165** that is configured as dovetail slot to movably or slidingly receive the second angled rear **149** of the second blade **137**, whereby the second angled upper slot **165** and the second angled rear **149** together define a dovetail joint. Other manners of connection may be used and are contemplated. The slant of the first angled upper slot **164**/second angled front **148** of the second blade **137** pair dovetail joint, and the slant of the second angled upper slot **165**/second angled rear **149** of the second blade **137** pair dovetail joint are opposite one another, whereby axial compression against the angled ends **148**/**149** of the second blade **137** causes, at least in part, the second blade **137** to extend radially outward from the distal end **122**. The pin **162** is connected to the distal end of the internal control rod. As the pin **162** is pulled axially towards the proximal end/end assembly **111**, the second angled slot **152** of the second blade **137** rides against the pin **162** during extension. Reverse axial movement of the internal control rod pushes the pin **162** against the second angled slot **152** such that the second blade **137** is retracted. The amount of axial movement in the proximal direction determines the amount of blade extension.

(48) FIGS. **22** and **23** show the extension of the first and second blades **136**, **137** from the distal end **122**, with one lateral side removed for clarity. In FIG. **22**, the first and second blades **136**, **137** are fully retracted. This is the position of the blades when the distal end **122** is inserted through an access tube (not shown). In FIG. **23**, the first and second blades **136**, **137** are fully extended. As the internal control rod is caused to axially move into the controller **113**, the pin **162**, being connected to the internal control rod, pulls the pin **162** and the nose **121** axially towards the controller **113**. This causes the first and second blades **136**, **137** to extend. The amount of extension is controlled by the controller **113**, and visually indicated by the indicator **124** on the longitudinal tube **119** of the shaft assembly **114**. Axial motion is signified by the double-headed arrow.

(49) In operation, the distal end or shaving head **122** of the vertebral endplate preparation tool with height adjustable (dynamic) blades **110** is inserted down an access tube. Once clear of the distal end of the access tube, the decorticating/shaving head **122** is inserted into the vertebral disc space. The blades are extended by rotation of the rotatable control knob **130** via the handle **112** to a particular radial height necessary for the particular disc space. Rotation of the blades then effects shaving of vertebral endplate material.

(50) Referring to FIG. **24**, there is depicted another form of a distal end/decorticating head **122** for the vertebral endplate preparation tool with height adjustable (dynamic) blades **110** dynamic shaver **110** of FIG. **15** designated **122***a*. The addition of the designation “a” after a callout number indicates an element or component that is different in some manner than the element or component of the vertebral endplate preparation tool with height adjustable blades **110** of FIGS. **15-23**. All other elements or components without the additional designation of “a”, or are not shown in FIG. **24**, are the same in form and function as the vertebral endplate preparation tool with height adjustable (dynamic) blades **110** of FIGS. **15-23**. Essentially, the shaving or decorticating head **122***a* has dovetail joints between the nose **121**, the distal end of the internal control rod of the shaft

assembly **114**, and the first and second blades **136***a*, **137***a* of the shaving/decorticating head **122***a* with a different profile than the dovetail joints between the nose **121**, the distal end of the internal control rod of the shaft assembly **114**, and the first and second blades **136**, **137** of the vertebral endplate preparation tool with height adjustable blades **110**. Thus, the first angled lower slot **166***a*, the second angled lower slot **167***a*, the first angled front **138***a*, and the first angled rear **139***a* form a dovetail joint that allows the ramped surfaces between them (which force extension) includes a side profile that allows the height of the first blade **136***a* to be controllably decreased when operated in the opposite direction (blade retraction direction). As well, the first angled upper slot **148***a*, the second angled upper slot **149***a*, the second angled front **148***a*, and the second angled rear **149***a* form a dovetail joint that allows the ramped surfaces between them (which force extension) includes a side profile that allows the height of the second blade **137***a* to be controllably decreased when operated in the opposite direction (blade retraction direction).

(51) Additionally, the first blade **136***a* has a pin slot **152***a* situated distally, and the second blade **137***a* has a pin slot (not seen in FIG. **24**) also situated distally. The pin **162***a* of the first and second lateral arms **158**, **170** is further situated distally. The first blade **136***a* has a middle opening **144***a*. The second blade **137***a* also has a middle opening **153***a*. Extension and retraction of the first and second blades **136***a*, **137***a* operates in the same manner as the vertebral endplate preparation tool with height adjustable blades **110**.

(52) Referring to FIGS. **25-27**, there is depicted a further form of a vertebral endplate preparation tool with height adjustable (dynamic) blades **110** having a different distal end/shaving/decorticating head designated **180**. The vertebral endplate preparation tool with height adjustable (dynamic) blades **110** of FIGS. **25-27** is different from the vertebral endplate preparation tool with height adjustable (dynamic) blades **110** of FIGS. **15-24** and the variation of the shaving head **22***a* of FIG. **24** because of the manner of how the shaving/decorticating head **80** operates, and thus configured. The shaving head **80** uses pins and slanted slots to effect extension and retraction rather than, or in addition to, dovetail joints. Other than the shaving head **80**, the vertebral endplate preparation tool with height adjustable blades **110** of FIGS. **25-27** has, at least generally, the same elements, components, and functionality as the vertebral endplate preparation tool with height adjustable blades **110** of FIGS. **15-24**, including the nose **121**, the first and second lateral arms **158**, **170** that extend from the nose **121** to the slots **156** and **168**, respectively of the longitudinal tube **119**, and connection pin **163**.

(53) The first blade **182** of the shaving head **180** has a first slanted rear **189** and a first slanted front **190**, a central opening **186**, a first angled rear slot **187**, and a first angled front slot **188**. The distal end of the longitudinal tube **119** has a first upper slanted channel **184** that receives the first slanted rear **189** of the first blade **182**, while the proximal end of the nose **121** has a second upper slanted channel **185** that receives the first slanted front **190** of the first blade **182**. A proximal guide pin **1100** extends from and through the first and second arms **158**, **170** and the first angled rear slot **187** of the first blade **182**. A distal guide pin **1101** extends from and through the first and second arms **158**, **170** and the first angled front slot **188** of the first blade **182**. The slant of the first angled front slot **188** and the slant of the first angled rear slot **187** are opposite.

(54) The second blade **183** of the shaving head **180** has a second slanted rear **197** and a second slanted front **198**, a central opening **194**, a second angled rear slot **195**, and a second angled front slot **196**. The distal end of the longitudinal tube **119** has a first lower slanted channel **191** that receives the second slanted rear **197** of the second blade **183**, while the proximal end of the nose **121** has a second lower slanted channel **192** that receives the second slanted front **198** of the second blade **183**. The proximal guide pin **1100** extends from and through the first and second arms **158**, **170** and the second angled rear slot **195** of the second blade **183**. The distal guide pin **1101** extends from and through the first and second arms **158**, **170** and the second angled front slot **196** of the second blade **183**. The second blade **183** also has a groove **199** for shaving and/or receiving vertebral endplate material. While not shown, the first blade **182** has a like groove. The slant of the

second angled front slot **196** and the slant of the second angled rear slot **195** are opposite.
(55) FIG. **25** depicts the first and second blades **182**, **183** in a retracted position wherein the first and second arms **158**, **170** are axially forward of the longitudinal tube **119**. FIGS. **26** and **27** depict the first and second blades **182**, **183** in an extended position wherein the first and second arms **158**, **170** are axially rearward. In this embodiment, when the proximal and distal guide pins **1100**, **1101** are caused to axially move through axial translation of the internal control rod (not seen) of the shaft assembly **114**, they pull (axial translation towards the proximal end/controller **113**) or push (axial translation towards the distal end/shaving head **122**) against the angled slots **187**, **188**, **195**, **196** causing the first and second blades to translate up (blade extension) or down (blade retraction). Axial movement is represented by the double-headed arrows.
(56) While the invention has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only a preferred embodiment has been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected. It should be understood that while the use of words such as preferable, preferably, preferred or more preferred utilized in the description above indicate that the feature so described may be more desirable, it nonetheless may not be necessary and embodiments lacking the same may be contemplated as within the scope of the invention.

Claims

1. A medical instrument for decorticating a vertebral endplate of a spine comprising: a handle; a controller operably connected to the handle and including a rotatable control knob with an internally threaded bore, wherein rotation of the handle rotates the rotatable control knob; a shaft assembly having a longitudinal tube defining a longitudinal tube proximal end and a longitudinal tube distal end with a control rod disposed therein, the control rod defining a distal control rod end and a proximal control rod, the proximal control rod end having external threading that is received in the internally threaded bore of the rotatable control knob for controlled axial movement of the control rod by the handle; a decorticating head disposed at the distal end of the longitudinal tube distal end and defining a decorticating head proximal end and a decorticating head distal end, the decorticating head distal end having a first front angled dovetail slot slanted at a first angle, and a second front angled dovetail slot slanted at a second angle opposite the first angle, the decorticating head proximal end having a first rear angled dovetail slot slanted at a third angle complementary to the first angle of the first front angled dovetail, and a second rear angled dovetail slot slanted at a fourth angle complementary to the second angle of the second front angled dovetail; a first blade having a first blade front angled dovetail slanted at a fifth angle being same as the first angle of the first front angled dovetail slot and a first blade rear angled dovetail slanted at a sixth angle being same as the third angle of the first rear angled dovetail slot, the first blade front angled dovetail situated in the first front angled dovetail slot of the decorticating head, and the first blade rear angled dovetail situated in the first rear angled dovetail slot of the decorticating head, whereby the first blade is radially moveable relative to the decorticating head; and a second blade having a second blade front angled dovetail slanted at a seventh angle being same as the second angle of the second front angled dovetail slot and a second blade rear angled dovetail slanted at an eighth angle being same as the fourth angle of the second rear angled dovetail slot, the second blade front angled dovetail situated in the second front angled dovetail slot of the decorticating head, and the second blade rear angled dovetail situated in the second rear angled dovetail slot of the decorticating head, wherein the second blade is radially moveable relative to the decorticating head; wherein axial force against the first blade via the control rod in a first axial direction effects radially outward movement of the first blade relative to the decorticating head via interaction of the first blade front angled dovetail and the first front angled dovetail slot of the decorticating head, and the first blade

rear angled dovetail situated in the first rear angled dovetail slot of the decorticating head, while axial force against the first blade via the control rod in a second axial direction opposite the first axial direction effects radially inward movement of the first blade relative to the decorticating head via interaction of the first blade front angled dovetail and the first front angled dovetail slot of the decorticating head, and the first blade rear dovetail situated in the first rear angled dovetail slot of the decorticating head; and wherein axial force against the second blade via the control rod in the first axial direction effects radially outward movement of the second blade relative to the decorticating head via interaction of the second blade front angled dovetail and the second front angled dovetail slot of the decorticating head, and the second blade rear angled dovetail situated in the second rear angled dovetail slot of the decorticating head, while axial force against the second blade via the control rod in the second axial direction opposite the first axial direction effects radially inward movement of the second blade relative to the decorticating head via interaction of the second blade front angled dovetail and the second front angled dovetail slot of the decorticating head, and the second blade rear angled dovetail situated in the second rear angled dovetail slot of the decorticating head.
2. The medical instrument of claim 1, wherein the first axial direction comprises compression against a front of the first blade and a front of the second blade.
3. The medical instrument of claim 2, wherein the first blade has a first angled cutout and a second angled cutout, and the second blade has a third angled cutout and a fourth angled cutout.
4. The medical instrument of claim 3, wherein the first angled cutout of the first blade is situated proximate to the decorticating head distal end, the second angled cutout of the first blade is proximate to the decorticating head proximal end, the third angled cutout of the second blade is situated proximate to the decorticating head distal end, and the fourth angled cutout of the second blade is proximate to the decorticating head proximal end.
5. The medical instrument of claim 4, wherein the first and second angled cutouts of the first blade are situated skewed relative to each other, and the third and fourth angled cutouts of the second blade are situated skewed relative to each other.